FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

HALF COURSE TEST – VII

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (08 – 11) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

   Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

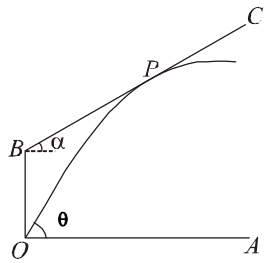
PART – I

SECTION – A

Straight Objective Type

*This section contains **7 multiple choice questions** numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct*

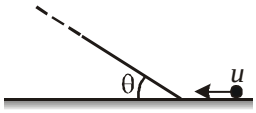
1. A particle is projected from a point O in a horizontal surface OA with speed v and angle of projection θ. It is just grazes the plane BC which make an angle α with the horizontal as shown in figure. The time taken by the projectile to reach P from the instant of projection is :



(A) $\frac{2u\sin\theta}{g\cos\alpha}$ (B) $\frac{u\sin\theta}{g\cos\alpha}$

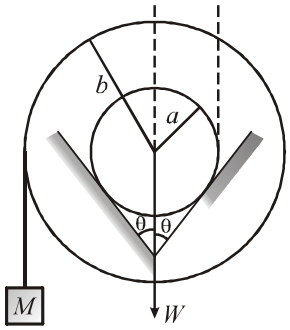
(C) $\frac{u\sin|\theta-\alpha|}{g\cos\alpha}$ (D) $\frac{u\cos|\theta-\alpha|}{g\cos\alpha}$

2. A particle of mass m is given initial speed u as shown in the figure. It moves to the fixed inclined plane without a jump, that is, its trajectory changes sharply from the horizontal line to the inclined line. All the surfaces are smooth and $90° \geq \theta > 0°$. Then the height to which the particle shall rise on the inclined plane (assume the length of the inclined plane to be very large)



(A) Increases with increase in θ
(B) decreases with increase in θ
(C) is independent of θ
(D) data insufficient

3. The axle of a wheel rests in V grooves as shown in figure and string, fixed to the rim of the wheel, supported a mass M. If μ is the coefficient of friction, W weight of wheel and axle, a is the radius of the axle and b the radius of the wheel. The least value of M which will cause motion is :



(A) $\frac{\mu Wa}{g(b(1+\mu^2)\sin\theta-\mu a)}$

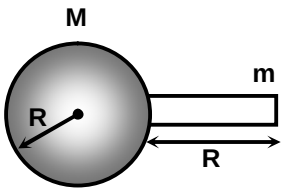
(B) $\frac{\mu Wb}{(a(1+\mu^2)\sin\theta-\mu b)g}$

(C) $\frac{\mu Wa}{(b(1+\mu^2)\sin\theta-\mu a)g}$

(D) $\frac{\mu Wb}{(a(1+\mu^2)\cos\theta-\mu b)g}$

Space for rough work

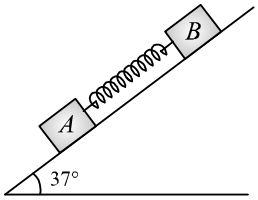
4. A uniform thin rod of mass m and length R is placed normally on surface of earth as shown. The mass of earth is M and its radius is R. Then the magnitude of gravitational force exerted by earth on the rod is:



(A) $\frac{GMm}{2R^2}$ (B) $\frac{GMm}{4R^2}$

(C) $\frac{4GMm}{9R^2}$ (D) $\frac{GMm}{8R^2}$

5. Two blocks A and B connected by spring (spring is in its natural length) are kept on smooth inclined plane and released. If acceleration of block B is 6 m/s^2 then that of block A at that instant (g = 10 m/s^2) :

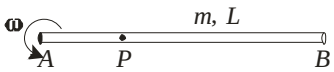


(A) must be greater than 6 m/s^2
(B) may be less than 6 m/s^2
(C) must be 6 m/s^2
(D) any thing possible depending on masses of blocks A & B

6. Heat is being supplied at a constant rate to a sphere of ice which is melting at the rate of 0.1 gm/sec. It melts completely in 100 sec. The rate of rise of temperature thereafter will be Lf = 80 cal/gm, S = 1 cal/gm-°C :

(A) 0.8 °C/sec. (B) 5.4 °C/sec.
(C) 3.6 °C/sec. (D) will change with time

7. A uniform thin, rod AB of length L and mass m is undergoing fixed axis rotation about end A, such that end A remains stationary as shown. The kinetic energy of section AP of rod is equal to kinetic energy of section BP of rod at an instant. Then the ratio of length AP and length AB, that is, $\frac{AP}{AB}$



(A) $\frac{1}{2}$ (B) $\frac{1}{2^{1/3}}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2\sqrt{2}}$

Space for rough work

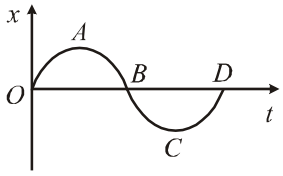
Multiple Correct Choice Type

*This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct*

8. Two particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm and 1 Hz respectively. Equilibrium positions of the particles A and B are at the co-ordinates (3, 0) and (0, 4) respectively. At t = 0, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin. Choose the correct option(s) :
(A) The distance between A and B is 3 cm at $t = t_0$ sec. It will again be 3 cm at $t = (t_0 + 1)$ sec.
(B) The distance between A and B can never be 5 cm.
(C) The maximum magnitude of force acting on each particle during the motion is same.
(D) Kinetic energy of the system (A and B) is conserved

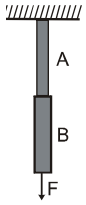
9. When the temperature of a copper coin is raised by 80°C, its diameter increases by 0.2%
(A) Percentage rise in the area of a face is 0.4%
(B) Percentage rise in the thickness is 0.4%
(C) Percentage rise in the volume is 0.6%
(D) Coefficient of linear expansion of copper is 0.25×10^{-4} C°$^{-1}$

10. A particle has a rectilinear motion and the figure gives its displacement as a function of time. Which of the following statements are true with respect to the motion



(A) in the motion between O and A the velocity is positive and acceleration is negative
(B) between A and B the velocity and acceleration are positive
(C) between B and C the velocity is negative and acceleration is positive
(D) between C and D the acceleration is positive

11. The wires A and B shown in the figure, are made of the same material and have radii of cross-section r_A and r_B. When a constant force F is applied as shown, one of the wires breaks. (Assume the m asses of the wire to be negligible)



(A) A will break before B if $r_A > r_B$
(B) B will break before A if $r_A < r_B$
(C) Either A or B will break if $r_A = r_B$
(D) The lengths of A and B must be known to decide which wire will break

Space for rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Questions 12 and 13

A cutting tool under microprocessor control has several forces acting on it. One force is $\vec{F} = \alpha xy^2\hat{j}$, a force in the negative y-direction whose magnitude depend on the position of the tool. The constant is α = 2.50 N. Consider the displacement of the tool from the origin to the point x = 3.00 m, y = 3.00 m.

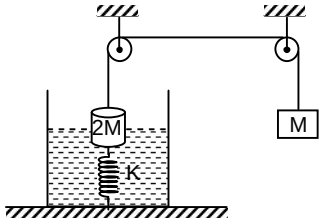
12. Calculate the work done on the tool by $\vec{F}$. If this displacement is along the straight line y = x that connects these two points ?
(A) 2.50 J (B) 500 J
(C) 50.6 J (D) 2 J

13. What can you predict about $\vec{F}$?
(A) Force is non-conservative
(B) Force is conservative
(C) Force is neither conservative nor non-conservative
(D) Data insufficient to conclude

Space for rough work

Paragraph for Question Nos. 14 to 16

A uniform cylindrical block of mass 2M and cross–sectional area A remains partially submerged in a non viscous liquid of density ρ, density of the material of the cylinder is 3ρ. The cylinder is connected to lower end of the tank by means of a light spring of spring constant K. The other end of the cylinder is connected to another block of mass M by means of a light inextensible string as shown in the figure. The pulleys shown are massless and frictionless and assume that the cross–section of the cylinder is very small in comparison to that of the tank. Under equilibrium conditions, half of the cylinder is submerged. [given $K > 8 A \rho g$]



14. Under equilibrium conditions

(A) extension of the spring is $\frac{Mg}{3K}$ (B) compression of the spring is $\frac{2Mg}{3K}$

(C) compression of the spring is $\frac{Mg}{6K}$ (D) extension of the spring is $\frac{Mg}{6K}$

15. By what maximum distance cylinder will be pushed downward into the liquid from equilibrium position, so that when it is set free then tension in the string will not vanish. [Assume at equilibrium position system was at rest]

(A) $\frac{3Mg}{(K + A\rho g)}$ (B) $\frac{3Mg}{2(K + A\rho g)}$

(C) $\frac{8Mg}{3(K + A\rho g)}$ (D) $\frac{3Mg}{(2K + 3A\rho g)}$

16. If the cylinder is pushed down from equilibrium by a distance which is half the distance as calculated in the above question determine time period of subsequent motion

(A) $2\pi\sqrt{\frac{3M}{2(K + A\rho g)}}$ (B) $2\pi\sqrt{\frac{M}{2K + A\rho g}}$

(C) $2\pi\sqrt{\frac{3M}{K + A\rho g}}$ (D) $2\pi\sqrt{\frac{2M}{K + 3A\rho g}}$

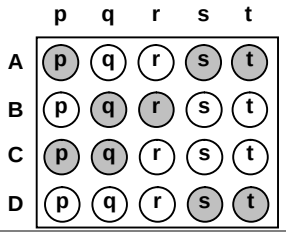
Space for rough work

SECTION – B

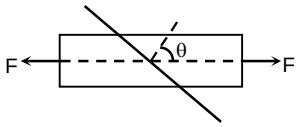
(Matrix-Match Type)

*This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.*

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:

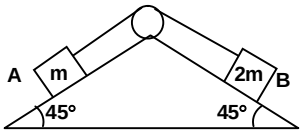


1. A bar of cross-section A is subjected two equal and opposite tensile forces at its ends consider a plane section of the bar whose normal makes an angle θ with the axis of the bar.



	Column-I	Column-II
(A)	What is the tensile stress on this plane ?	(p) $\theta = 45°$
(B)	What is the shearing stress on this plane ?	(q) $\theta = 0°$
(C)	For what value of θ is the tensile stress is maximum?	(r) $(F/2A) \sin 2\theta$
(D)	For what value of θ is the shearing stress maximum?	(s) $(F/A) \cos^2\theta$
		(t) $\theta = 60°$

2. Two blocks A and B of mass m and 2m are placed on a fixed triangular wedge are placed on a fixed triangular wedge by massless inextensible string as shown. The pulley is massless and frictionless. The coefficient of friction between block A and the wedge is 2/3 and that between B and the wedge is 1/3. (Take m = 1 kg, g = 10 m/s^2)



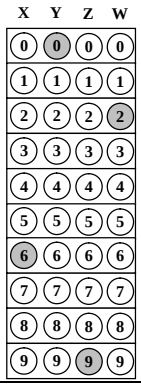
	Column – I		Column – II
(A)	Friction force between block A and wedge	(p)	$\frac{5}{\sqrt{2}}$ units
(B)	Friction fore between block B and wedge	(q)	$\frac{10}{3\sqrt{2}}$ units
(C)	Tension in the string	(r)	$\frac{10\sqrt{2}}{3}$ units
(D)	Maximum friction force between Block B and wedge.	(s)	$30\sqrt{2}$ units
		(t)	$\frac{20\sqrt{2}}{3}$ units

Space for rough work

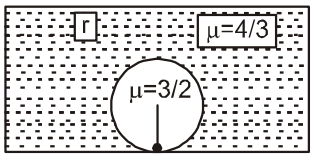
SECTION – C

Integer Answer Type

*This section contains **6 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown*



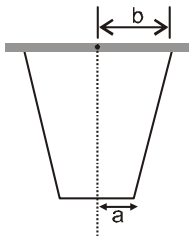
1. A glass sphere having refractive index (3/2) is having a small irregularity at its centre. It is placed in a liquid of refractive index 4/3 such that surface of liquid is r high above sphere where r is radius of sphere. If irregularity is viewed from above normally, calculate distance in cm from centre where eye will observe the irregularity is 10P in cm. Find the value of P.(r = 20 cm)



2. Two block of mass m and M are connected means of a metal of a metal wire passing over a frictionless fixed pulley. The area of cross-section of the wire is 6.67×10^{-9} m^2 and its breaking stress is 2×10^9 Nm^{-2}. If m = 1kg. Find the maximum value of M in kg for which the wire will not break. ($g=10$ m/s^2).

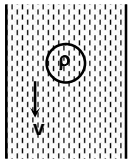
Space for rough work

3. A body of mass 3.14 kg is suspended from one end of a wire of length 10m. The radius of the wire is changing uniformly from 9.8×10^{-4} m at one end to 5×10^{-4} m at the other end. Find the change in length of the wire in mm. Young's modulus of the material of the wire is 2×10^{11} N/m^2. (mass of wire is negligible).

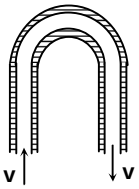


4. A conducting rod of length 1 m aligned along the unit vector $\left(\frac{3}{5}\hat{j}+\frac{4}{5}\hat{k}\right)$, is moving with a velocity $\vec{v}=\left(2\hat{i}\right)$ m/s. The motion take place in a uniform magnetic field $\vec{B}=\left(\hat{j}-2\hat{k}\right)$ tesla. Find the magnitude of induced emf in across the rod in volt.

5. A liquid of density ρ = 1000 kg/m^3 viscosity η = 0.1 Ns/m^2 is flowing down in vertical pipe of large cross section. A small ball of density ρ_0 = 100 kg/m^3 and r = 5 cm will be at rest in flowing liquid, if velocity of flowing liquid is 10k m/s. Then find the value of k.



6. A pipe of circular cross-section of inner radius r and outer radius 2r is bent into a semi circle as shown in the diagram. A fluid of density $\rho = 10^3$ kg/m^3 is flowing through it. The breaking stress of the material of the pipe is 3×10^7 N/m^2. The maximum velocity with which the fluid can flow in the pipe is k $\times$ 100 m/s. Find value of k.



Space for rough work

Chemistry

PART – II

SECTION – A

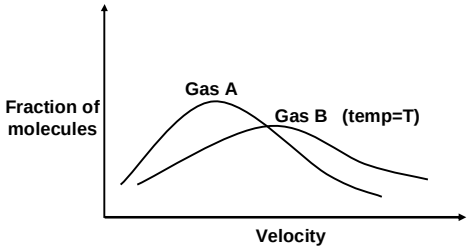
Straight Objective Type

*This section contains **7 multiple choice questions** numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct*

1. $Na_2SO_3 + S \xrightarrow[\Delta]{\text{alkaline}}$
   (A) Na_2SO_3 (B) $Na_2S_2O_3$
   (C) Na_2SO_4 (D) Na_2S

2. How much energy is required for breaking a hydrogen bond?
   (A) 2 kcal (B) 100 kcal
   (C) 2 kJ (D) 200 kJ

3. Which of the following statement is incorrect for the given graph?



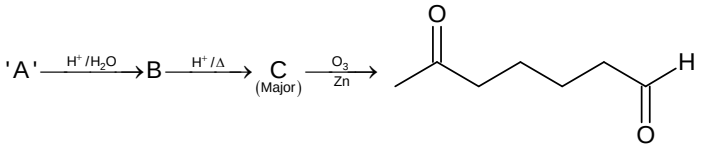
   (A) The molecular weight of gas A will be greater than gas B
   (B) The average kinetic energy of both the gases will be similar at a particular temperature
   (C) The average velocity of particles of gas A will always be less than of particles of gas B at temperature T.
   (D) The velocity of any particle of gas A can not be greater than velocity of any particle of gas B at same temperature.

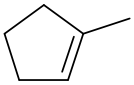
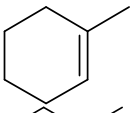
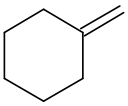
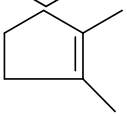
4. A water sample is found to have 384 ppm of SO_4^{2-}. What will be the degree of hardness of water sample?
   (A) 200 ppm (B) 300 ppm
   (C) 400 ppm (D) 100 ppm

Space for rough work

5. How many structural isomers are possible with molecular formula C_7H_{16}?
(A) 6 (B) 8
(C) 10 (D) 12

6. Which of the following is mismatched?
(A) $LiCO_3 \Rightarrow$ least stable carbonate among alkali metal carbonates.
(B) $LiF \Rightarrow$ almost insoluble in water due to its high covalent character.
(C) $Ba(OH)_2 \Rightarrow$ most soluble hydroxide among alkaline earth metal hydroxide.
(D) $BeSO_4 \Rightarrow$ more soluble than $BaSO_4$ in water.

7. 

(A) and (C) are isomers.
Which of the following can be structure of (A)?
(A) 
(B) 
(C) 
(D) 

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

8. An electron in a hydrogen gas sample is excited to 5th excited level. Which of the following will be true for its spectrum?
(A) In spectrum total lines possible for Balmer series will be one
(B) In spectrum total lines possible for Lymann series will be five
(C) In spectrum total lines possible for IR region will be six
(D) In spectrum total lines possible for UV region will be five

Space for rough work

9. Which of the following relation is/are correct?

(A) $PV = \frac{RT}{d}$ (B) $d = \frac{PM}{RT}$

(C) $\frac{PV^2Md}{RT} = w^2$ (D) $n = \frac{w^2}{VMd}$

Where P = pressure (in atm)
V = volume (in litre)
T = temperature (in K)
R = Gas constant (0.0821 atm $LK^{-1}mol^{-1}$)
n = mole
d = density (g/L)
M = molecular weight (in g)

10. Which of the following oxides are basic only?

(A) MgO (B) BeO
(C) Al_2O_3 (D) BaO

11. Which bond properties are consistent with one another?

	Bond order	Bond length	Vibrational frequency
(A)	Higher	Longer	Lower
(B)	Higher	Shorter	Higher
(C)	Lower	Longer	Lower
(D)	Lower	Shorter	Higher

Comprehension Type

*This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct*

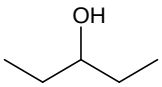
Paragraph for Question Nos. 12 to 13

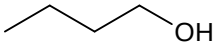
Among various reaction intermediates one of the important intermediate is carbocation in which carbon is bearing positive charge and is sp^2 hybridized so its structure is planar.

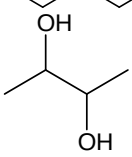
Many reactants like alkyl halides, alcohols and alkanes show reactions via carbonation. Higher the stability of respective intermediate higher the reactivity of the reactant towards that reaction.
Carbocations are stabilized mainly by aromaticity, resonance, hyperconjugation and positive inductive effect.

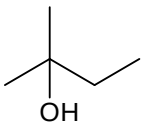
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12. Which of the following is the most reactive among given alcohols towards dehydration?

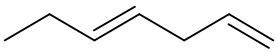
(A) 

(B) 

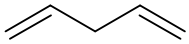
(C) 

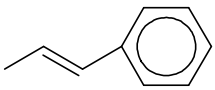
(D) 

13. Which of the following alkenes will be undergoing acid – catalysed hydration fastest?

(A) 

(B) 

(C) 

(D) 

Paragraph for Question Nos. 14 to 16

According to VSEPR theory the repulsion between various lone pairs & bond pairs are
lp – lp > lp – bp > bp – bp
Due to the repulsion the shape of the molecules also gets distorted from the normal geometries.

14. Which of the following molecule is having unsymmetrical shape?
(A) SF_4 (B) SF_6
(C) XeF_2 (D) XeF_4

15. Which of the following molecule has only bp – lp repulsion & bp – bp repulsion?
(A) CO_2 (B) CH_4
(C) $XeOF_4$ (D) XeF_2

16. Which of the following molecule is not unsymmetrical despite of being presence of lone pairs on central atom?
(A) CS_2 (B) SF_6
(C) ClF_3 (D) I_3^-

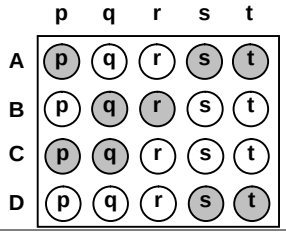
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SECTION – B

(Matrix-Match Type)

*This section contains **2 questions.** Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.*

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:



1. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	$H_2(g)+I_2(g)\rightleftharpoons 2HI(g)$	(p)	Change in equilibrium constant on increasing temperature
(B)	$PCl_5(g)\rightleftharpoons PCl_3(g)+Cl_2(g)$	(q)	Equilibrium is unaffected by adding inert gas at constant volume
(C)	$N_2(g)+3H_2(g)\rightleftharpoons 2NH_3(g)$	(r)	Equilibrium is unaffected by adding inert gas at constant pressure
(D)	$CaCO_3(s)\rightleftharpoons CaO(s)+CO_2(g)$	(s)	Equilibrium shift to forward on increasing pressure
		(t)	Increase in reactant concentration shifts equilibrium in forward direction

2. Match the Column – I with Column – II:

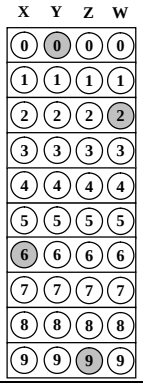
Column – I		Column – II	
(A)	C_3H_6O	(p)	Can exhibit functional isomerism (considering no cyclic structure)
(B)	C_4H_{10}	(q)	Can exhibit optical activity
(C)	C_7H_{16}	(r)	Can exhibit position isomerism
(D)	C_2H_7N	(s)	Can exhibit metamerism
		(t)	Can exhibit chain isomerism

Space for rough work

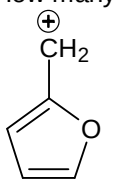
SECTION – C

Integer Answer Type

*This section contains **6 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown*



1. How many significant resonating structures are possible for the given species including it?



2. The sum of oxidation states of sulphur in following compounds will be.
Cu_2S, FeS_2, $Na_2S_2O_3$, $Na_2S_2O_4$

3. The number of possible alcohols (saturated) with different boiling point with molecular formula C_4H_8O will be.

4. 2 mol of a solution containing A^{n+} requires 0.8 mol of $M_nO_4^-$ for the oxidation of A^{n+} to AO_3^- in acidic medium. The value of n is. (M is any metal)

5. 2 moles of A are mixed with 4 moles of B. When 2 moles of C and 1 mole of D is formed according to reaction. The value of equilibrium constant is $A + 2B \rightleftarrows 2C + D$.

6. When an electron jumps from 6th excited level in a single H–atom. How many maximum lines can be obtained in Lyman series in the spectrum?

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

*This section contains **7 multiple choice questions** numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct*

1. Which of there represent a triangle
   (A) $|z-1| = |z-2|$ (B) $|z-1| = |z-2| = |z-i|$
   (C) $|z-1| - |z-2| = 2a$ (D) $|z-1|^2 + |z-2|^2 = 4$.

2. If α, β, γ are the cube roots of p(p < 0), then for any x, y, z, $\frac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha} =$
   (A) $\frac{1}{2}\left(-1-i\sqrt{3}\right)$ (B) $\frac{1}{2}\left(1+i\sqrt{3}\right)$
   (C) $\frac{1}{2}\left(1-i\sqrt{3}\right)$ (D) none of these.

3. If A_1, A_2; G_1, G_2; and H_1, H_2 be two AMs, GMs and HMs between two numbers then $\frac{G_1G_2}{H_1H_2} \times \frac{H_1+H_2}{A_1+A_2} =$
   (A) 1 (B) 0
   (C) 2 (D) 3.

4. If a line intersect a hyperbola at (– 2, – 6) and (4, 2) and one of the asymptote at (1, – 2), then the centre of the hyperbola is
   (A) (7, 6) (B) (1, – 2)
   (C) (10, 10) (D) (– 5, – 10)

Space for rough work

5. If $0 \le A, B, C \le \pi$ and $A + B + C = \pi$ than the minimum value of sin3A + sin3B + sin3C is

(A) – 2 (B) $\frac{-3\sqrt{3}}{2}$

(C) 0 (D) none of these

6. A seven digit number without repetition and divisible by 9 is to be formed by using 7 digits out of 1, 2, 3, 4, 5, 6, 7, 8, 9. The number of ways in which this can be done is;

(A) 9! (B) 2 (7!)

(C) 4 (7!) (D) none of these

7. The largest two digit prime factor of $^{200}C_{100}$ is

(A) 61 (B) 59

(C) 17 (D) none of these

Multiple Correct Choice Type

*This section contains **4 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct*

8. If $ac > b^2$ then the sum of the coefficients in the expanssion of $(a\alpha^2x^2 + 2b\alpha x + c)^n$; (a, b, c, $\alpha \in R$, $n \in N$) is

(A) positive if a> 0 (B) positive if c > 0

(C) negative if a < 0 , n is odd (D) positive if c < 0, n is even.

9. If f(x) and g(x) are two polynomials such that the polynomial $h(x) = x f(x^3) + x^2 g(x^6)$ is divisible by $x^2 + x + 1$, then

(A) f(1) = g(1) (B) f(1) = - g(1)

(C) $f(1) = g(1) \neq 0$ (D) $f(1) = -g(1) \neq 0$

10. If three positive unequal numbers a, b, c are in H. P., then

(A) a + c > 2b (B) $a^2 + c^2 > 2b^2$

(C) $a^2 + c^2 > 2ac$ (D) none of these.

11. If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $x \in [0, \pi]$, then

(A) $x = \pi/4$ (B) y =0

(C) y = 1 (D) $x = 3\pi/4$.

Space for rough work

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraphs 2 multiple choice questions and based on the other paragraph 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Read the following writeup carefully:

$y = ax^2 + bx + c = 0$ is a quadratic equation which has real roots if and only if $b^2 - 4ac \geq 0$. If $f(x, y) = 0$ is a second degree equation, then using above fact we can get the range of x and y by treating it as quadratic equation in y or x. Similarly $ax^2 + bx + c \geq 0 \ \forall \ x \in R$ if $a > 0$ and $b^2 - 4ac \leq 0$.

Now answer the following questions:

12. If $0 < \alpha, \beta < 2\pi$, then the number of ordered pairs (α, β) satisfying $\sin^2(\alpha + \beta) - 2\sin\alpha \sin(\alpha + \beta) + \sin^2\alpha + \cos^2\beta = 0$ is
    (A) 2 (B) 0
    (C) 4 (D) none of these

13. If $A + B + C = \pi$, then the maximum value of $\cos A + \cos B + k\cos C$ (where $k > 1/2$) is
    (A) $\frac{1}{k} + \frac{k}{2}$ (B) $\frac{2k^2 + 1}{3}$
    (C) $\frac{k^2 + 2}{2}$ (D) $\frac{1}{2k} + k$

Space for rough work

Paragraph for Question Nos. 14 to 16

Read the following comprehension carefully:

a, b, c are complex numbers and all roots of $z^3 + az^2 + bz + c = 0$ satisfy $|z| = 1$.

Now answer the following questions.

14. The value of $|a| - |b|$ is
(A) 0 (B) 1
(C) – 1 (D) 2

15. The number of distinct real roots of the equation $z^3 + |a|z^2 + |b|z + |c| = 0$ is
(A) 1 (B) 2
(C) 3 (D) can't say

16. All the roots of $z^3 + |a|z^2 + |b|z + |c| = 0$ satisfy
(A) $|z| < 1$ (B) $|z| = 1$
(C) $|z| > 1$ (D) can't say

Space for rough work

SECTION – B

(Matching List Type)

*This section contains **2 multiple choice questions**. Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct*

1. Match the following Column-I with Column-II

	Column – I		Column – II
(A)	Normal of parabola $y^2 = 4x$ at P and Q meets at R $(x_2, 0)$ and tangents at P and Q meets at T $(x_1, 0)$, then if $x_2 = 3$, then the area of quadrilateral PTQR is	(p)	3
(B)	The length of latus rectum plus tangent PT will be	(q)	6
(C)	The quadrilateral PTQR can be inscribed in a circle, then the value of circumference/ 4π will be	(r)	1
(D)	The number of normals that can be drawn to the parabola from $R(x_2, 0)$ is	(s)	8
		(t)	5

2. Match the following Column-I with Column-II

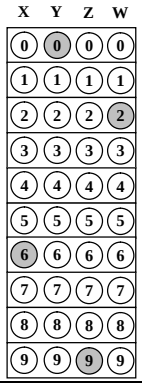
	Column – I		Column – II
(A)	In an A.P. $S_p = a$, $S_q = b$, $S_r = c$, then $\frac{a}{p}(q-r)+\frac{b}{q}(r-p)+\frac{c}{r}(p-q)$ is	(p)	3
(B)	In a G.P., $T_r = R$, $T_s = S$, $T_t = T$, then $R^{s-t}\cdot S^{t-r}\cdot T^{r-s}$ is	(q)	2
(C)	If x, y, z are both in A.P. and G.P., $T_m = x$, $T_n = y$, $T_p = z$, then $x^{y-z}\cdot y^{z-x}\cdot z^{x-y}$ is	(r)	1
(D)	If a, b, c, are both in G.P. and H.P., $T_p = 0$, $T_q = b$, $T_r = c$, then $a(b-c)\log a + b(c-a)\log b + c(a-b)\log c$	(s)	0
		(t)	4

Space for rough work

SECTION – C

Integer Answer Type

*This section contains **6 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown*



1. If the roots of $x^3 - 12x^2 + 39x - 28 = 0$ are in A.P. then mod of their common difference will be …

2. If the roots of the equation $x^2 - bx + c = 0$ and $x^2 - cx + b = 0$ $[b \neq c]$ differ by the same quantity then $-(b + c)$ is …..

3. $\frac{c_1}{c_0} + 2\frac{c_2}{c_1} + 3\frac{c_3}{c_2} + ... + 15\frac{c_{15}}{c_{14}} = 5!k$ then k is ….

4. Value of '(–k)' such that the lines $2x - 3y + k = 0$, $3x - 4y - 13 = 0$ and $8x - 11y - 33 = 0$ are concurrent is …..

5. No. of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is …..

6. If the sum of the coefficients in the expansion of $(\alpha x^2 - 2x + 1)^{35}$ is equal to the sum of the coefficients in the expansion of $(x - \alpha y)^{35}$, then α = ……..

Space for rough work